Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/04/2005
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Role of the Interleukin-2 Receptor in Signal Transduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NON-PROFIT	**Phone:**	415-750-6940
Organization Name:	Northern California Institute for Research and Education		
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PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	01/15/1999	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cogoli, Augusto (Swiss Federal Institute of Technology)		
Grant/Contract No.:	NAG2-1286		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This work is collaboration between Professor Augusto Cogoli and Professor Millie Hughes-Fulford to study the role of the interleukin-2 receptor in signal transduction of T cells. The nearly total loss of T cell activation in real microgravity was discovered in a Spacelab in 1983 and was confirmed later by several experiments in space. Inhibition of IL-2R expression is probably one of the causes of such loss of activation. It is planned to analyze the pathway of mitogenic signal transduction in T cells in true microgravity as well as at variable G levels between 0 and 1xg. The objectives of this research project are: 1. To investigate selected critical steps of T cell activation 2. To test the hypothesis that a failure in the expression of the IL-2R is causing the loss of activity in microgravity 3. To identify windows of sensitivity to gravity during the activation process 4. To establish thresholds of gravi-sensing T cells. The main experimental approach will consist of the activation of cultures of purified peripheral blood cells with T cell		

	mitogens in real and simulated microgravity. The specific IL-2R mRNA will be quantitatively determined with the reverse Transcriptase-polymerase chain reaction (RT-PCR) technology, the insertion of IL-2R in the membrane will be visualized by immunofluorscence and its secretion in the supernatant will be measured by immunoassay. Signal transduction will be furthered analyzed by determining the early expression of oncogenes. The Laboratory of Cell Growth (LCG), headed by Dr. Millie Hughes-Fulford is responsible for analysis of gene expression in T cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All terrestrial life began in a gravity field. This study of alterations of early T-cell activation microgravity gives us the unique opportunity to examine the role of Earth's gravity in immune function. In the late 60's returning Apollo astronauts were found to have reduced immune function after spaceflight, taking approximately 7 days to recover normal function. Later studies by Dr. Cogoli demonstrated changes in the T-cells during exposure to microgravity. This program will examine the expression of IL-2, and the three IL-2 receptor subunits (alpha, beta and gamma) in microgravity with and without an artificial 1g gravity vector and compare those results with ground controls. These studies will show the regulation of the earliest signals that cause the T-cell to activate and the role of normal earth's gravity in that signaling.
Task Progress:	With the loss of Columbia and her crew, as well as our samples, Augusto Cogoli and I analyzed our possibilities and continued our objectives here on the ground by altering gravity in the random positioning machine (RPM) during T cell activation. We have completed ground based studies using the gene chip and have a manuscript in press in Cellular Signalling which describes for the first time the role of PKA pathway, cAMP and AP1 in early T cell activation. The work on the RPM is submitted for consideration. We are continuing the work to study early signalling pathways at the protein level.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Millie Hughes-Fulford, "Signal transduction and mechanical stress. " Science’s STKE. 2004 Aug 31;2004 (249):RE12 , Aug-2004
Articles in Peer-reviewed Journals	Millie Hughes-Fulford, Eiko Sugano Thomas Schopper Chai-Fei Li J. B. Boonyaratanakornkit and Augusto Cogoli "Early immune response and regulation of IL-2 receptor subunits" Cellular Signalling In Press 2005 , Feb-2005
Articles in Peer-reviewed Journals	Millie Hughes-Fulford "Function of the cytoskeleton in gravisensing during spaceflight." Adv Space Res. 2003;32(8):1585-93 , Oct-2004
Articles in Peer-reviewed Journals	Hatton JP, Pooran, M. C-F Li, C Luzzio, Fulford, MH "A short pulse of mechanical force induces gene expression and growth in MC3T3-E1 osteoblasts via an ERK 1/2 pathway" JBMR Jan;18(1):58-66, Jan-2003 , Jan-2003
Books/Book Chapters	M. Hughes-Fulford "Physiological effects of microgravity on osteoblast morphology and cell biology" in "Advances in space Biology and Medicine" Ed. A. Cogoli Elsevier , 32(8):1585-93 , Jan-2003